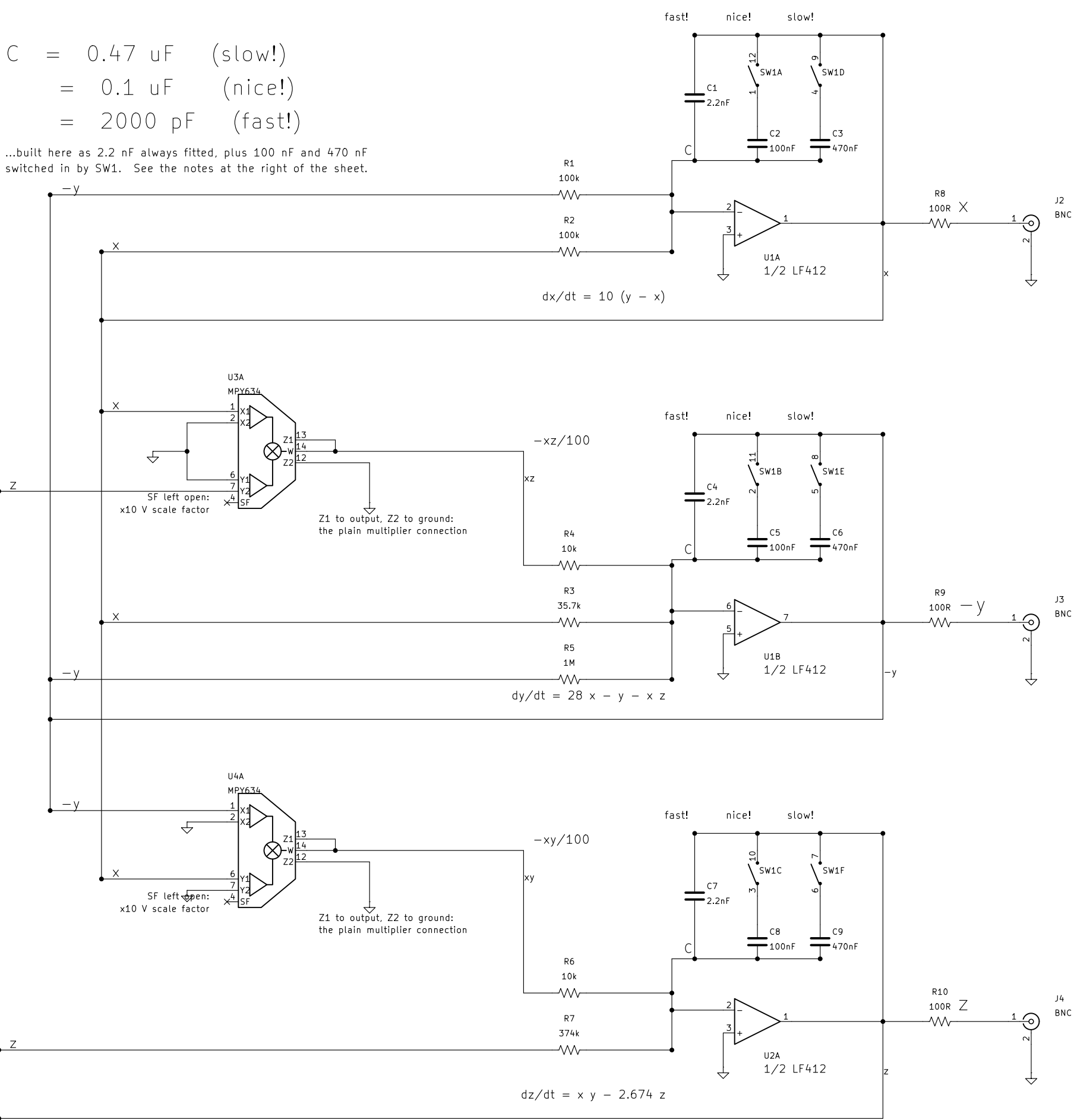
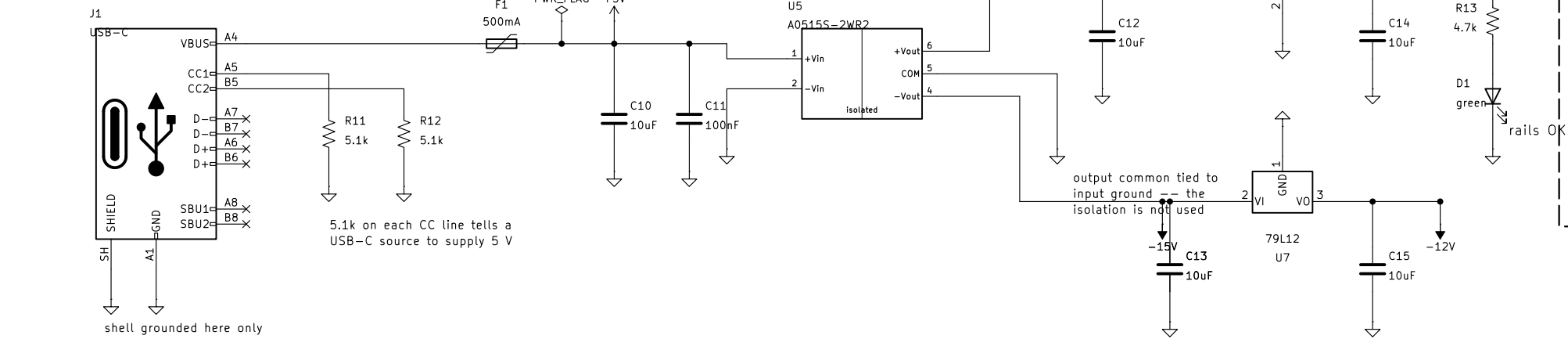


C = 0.47 uF (slow!)
= 0.1 uF (nice!)
= 2000 pF (fast!)

...built here as 2.2 nF always fitted, plus 100 nF and 470 nF switched in by SW1. See the notes at the right of the sheet.

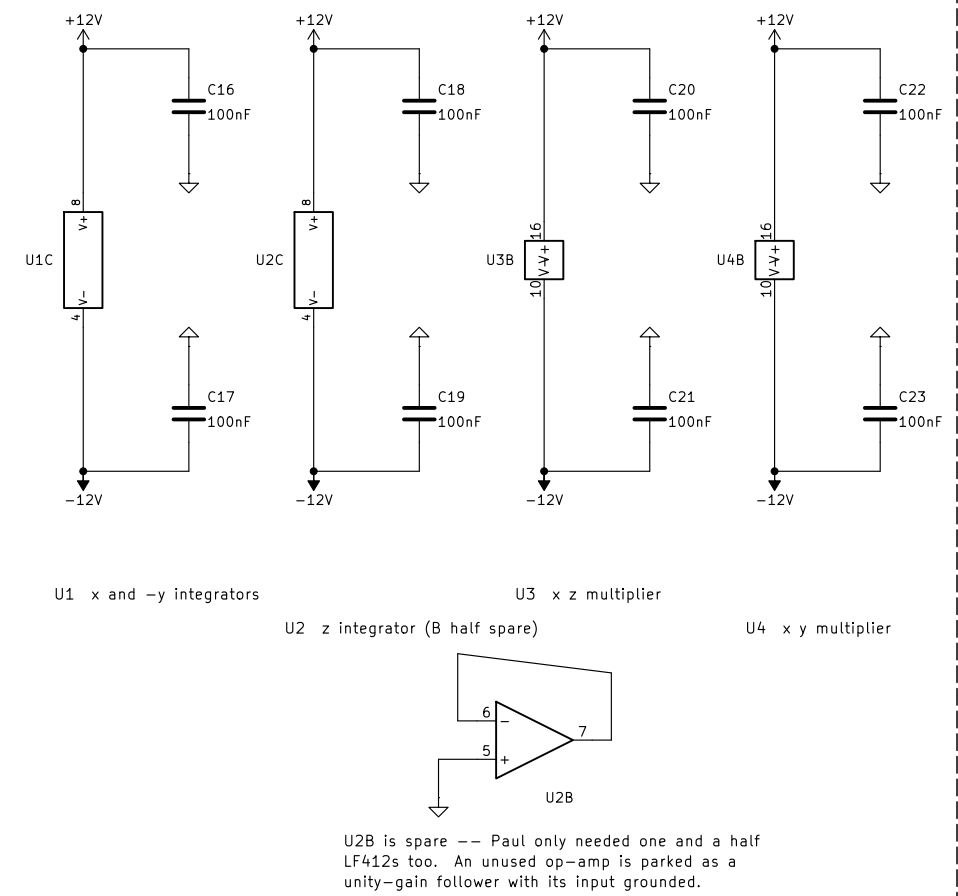


POWER: USB-C 5 V in, +/-12 V out
about 20 mA per rail; the 2 W module is rated +/-66 mA

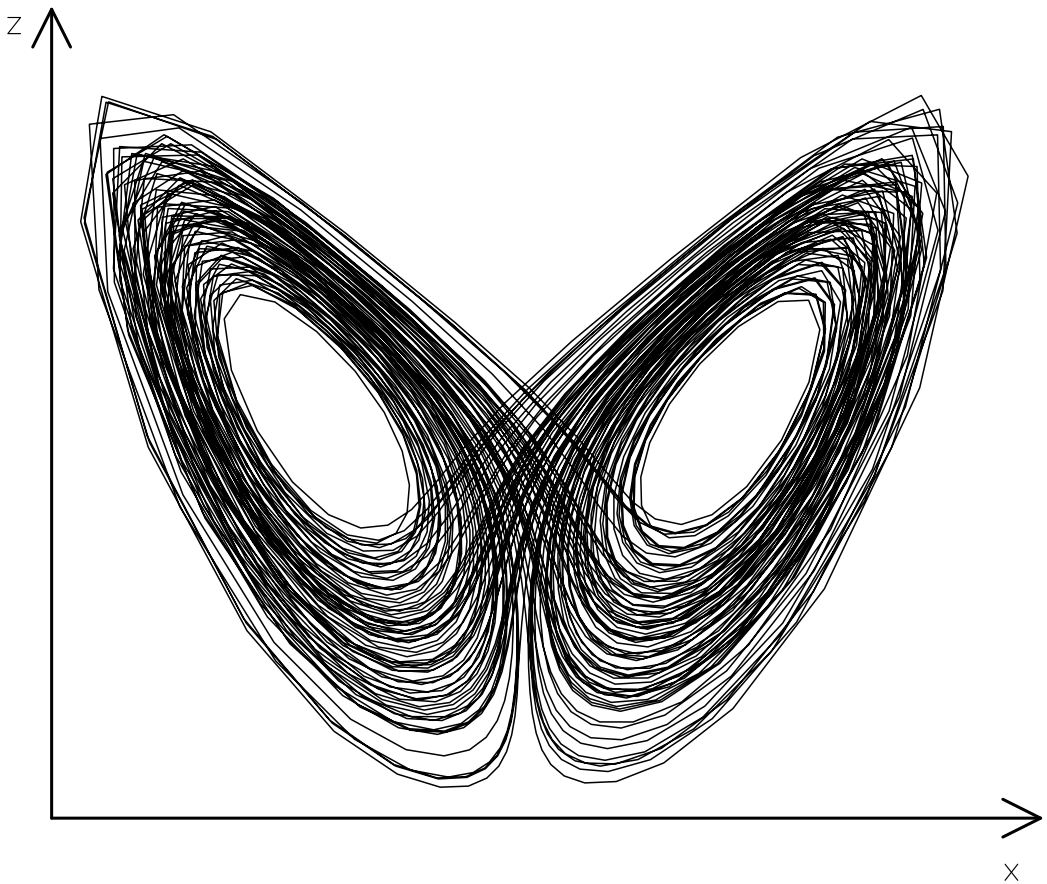


SUPPLY PINS AND BYPASSING

one 0.1 uF at every supply pin, right at the package



Hook x and z to a scope in X-Y and this is what you get: the attractor's characteristic owl's face. The trace below is the real solution of these equations, integrated at s = 10, r = 28, b = 8/3.



HOW THIS CIRCUIT SOLVES THE EQUATIONS

$$\begin{aligned} dx/dt &= s(y - x) & s &= 10 \\ dy/dt &= r x - y - x z & r &= 28 \\ dz/dt &= x y - b z & b &= 8/3 \end{aligned}$$

Each op-amp is an integrator with the terms of one derivative summed at its inverting input, so the weight of every term is simply 1 MEG / R :

$$\begin{aligned} R2 = R1 = 100k &\rightarrow 1M/100k = 10.00 = s \quad (\text{and the } -x \text{ damping}) \\ R3 = 35.7k &\rightarrow 1M/35.7k = 28.01 = r \\ R5 = 1M &\rightarrow 1M/1M = 1.00 = \text{the } -y \text{ term} \\ R7 = 374k &\rightarrow 1M/374k = 2.674 = b \quad (8/3 = 2.667, 0.3 \% \text{ high}) \\ R4 = R6 = 10k &\rightarrow 1M/10k = 100 \quad \text{cancels the multipliers' } /100 \end{aligned}$$

Voltagess carry the dimensionless variables at 0.1 V per unit, so x swings +/-2 V, y +/-2.7 V and z runs 0 to 4.8 V. An MPY634 forms A*B/10, so the product of two 0.1 V/unit signals comes out 100x too small -- which is exactly what the 10k weighting resistors R4 and R6 put back. The minus signs on both products are free: U3 and U4 simply have their differential inputs swapped.

The time scale is tau = 1 MEG x C, and C is switchable:

$$\begin{aligned} \text{SW1-1,2,3 off} \quad \text{SW1-4,5,6 off} \quad C &= 2.2 \text{ nF} \quad \text{fast!} \quad \sim 3 \text{ ms per lobe} \\ \text{SW1-1,2,3 ON} \quad \text{SW1-4,5,6 off} \quad C &= 102 \text{ nF} \quad \text{nice!} \quad \sim 0.15 \text{ s per lobe} \\ \text{SW1-1,2,3 off} \quad \text{SW1-4,5,6 ON} \quad C &= 472 \text{ nF} \quad \text{slow!} \quad \sim 0.7 \text{ s per lobe} \end{aligned}$$

Paul's original ran on a bench +/-15 V supply and used 2000 pF, 0.1 uF and 0.47 uF. Here the three values are built up in parallel because no supplier stocks a 3-pole 3-position switch; 2.2 nF is always fitted.

MOUNTING

Four M3 holes, one per board corner. No electrical function, but with three BNCs along one edge the cables lever on the board, so it wants standoffs rather than a bare bench.

